GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 01

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Topics to be Covered

Topic-1 Calendar

Almanac

Topic-2 Clocks

CALENDAR

English → Solar

+1

Georgian

~~+12~~

Basic Questions

2023
2024

1 B.C
1 A.D

Date → Day?

01-01-01
00-00-00

1 Monday
0 Sundays

1. Use of CALENDAR?

2. How many months in a year consist ~~of 30 days~~ only?

11 Months

April
June
Sep
Nov

3. Which is the first day of a week?

SUNDAY

4. What is the difference between A.D. & B.C.?

Holy Year

Anno Domini

1632 B.C

Before Christ

LEAP YEAR Occurrence

400

1632 ✓ Leap Year

1712 → ✓

1291 ✗

1744 ✓

29th Feb

?

1796

1800 ✗

✗ ←

29th Feb ← 1804

Maximum → 8 years

29th Feb - 1996

29th Feb 2000

Minimum → 1 year

To be Noted:

365 days 5 hrs 32 min — secs

$365\frac{1}{4}$ days
OR
365 days 6 hrs

✓If a year is divisible by 4 then it's a leap year else normal. e.g. 1652, 1212, 1496, 1708, etc. are leap years whereas 1714, 1446, 2006 etc. are normal years.

✓If a century is divisible by 400 then it's a leap year else normal. e.g. 1600, 1200, 2000, 800 etc. are leap years whereas 1000, 1500, 2100 etc. are normal years.

6 hrs
6 hrs
6 hrs
6 hrs → 24 hrs → 1 day

1st year → 365
2nd year → 365
3 → 365
L.Y ← 366 days → 365 + 1

2004
2008
2012
2016
2020
2024
2028
2032
⋮
98

ODD DAYS

As today is 3rd June, 2024. The day is Monday.

✓What would be the day on 13th June, 2024?

10 days

40

+3

7 days

30 days

7) 30 (4 weeks)
28

2 → odd day

Thursday

60

ODD DAYS

✓All those number of days which can't be kept in a group of a week.

OR

✓ When given number of days is divided by 7, the remainder is called as odd days.

$$7\overline{)50}$$

Normal Year & Leap Year

$7 \overline{)365}$ (52 weeks)

35

15

14

1

N.Y. ⇒ 365 days ⇒ 1 odd day

L.Y. ⇒ 366 days ⇒ 2 odd days

First Century

04
08
12
16
20
24
28
32
...
96

×3

×7

×24

ODD DAYS → 0 - 6

100 years → 5 odd days

L.Y | N.Y.

24 | 76

×2 | ×1

48 | 76

6 + 6 = 12

Odd Days in Centuries

1700 | 1800 | 1900 | 2000 | 2100 | 2200 | 2300 | 2400 | 2500 | 2600 | 2700 | 2800 | 2900 | 3000

1700 × 100 → 5 odd days

1800 × 200 → 3 odd days

1900 × 300 → 1 odd day

2000 × 400 → ~~6~~ + 1 = 7

0 odd days

Questionnaire:

0 → Sunday
1 → Monday
2 → Tuesday
3 → Wednesday
4 → Thursday
5 → Friday
6 → Saturday

04

92

(31) J → 3

23 (8) F → 1

4

94 → 5 odd days

L.Y

N.Y.

23

71

4 46

1 71

8th Feb, 1995 → 3 odd days → Wednesday

1900 → 1 odd days

1994 → 6 odd

10 → 3 odd days

Questionnaire:

3rd June, 1947

(A) Tuesday

(B) Saturday

(C) Friday

(D) Monday

04

XII

44

1900 → 1 odd day

46 → 1

L.Y. N.Y.

22 35

1 + 0

2+0 = 2 odd day

1946 → 2 odd days

J – 3 A → 2

F – 0 M → 3

M – 3 J → 3

+ 0

Questionnaire:

#Q. Guru Nanak was born on 15th April, 1469. What was the week day?

(1) Saturday

(2) Tuesday

(3) Thursday

(4) None of these

$4 + 0 =$ 4 odd days

1400 → 3 odd days

68 → 1

L.Y 17 → 34

N.Y 51 → 51

$0 + 2 = 8$

1468 → 4 odd days

J - 3

F - 0

M - 3

A → 1

Wednesday

31st Dec, 1947 → 3

04

×11

44

1900 → 1

47 → 2

L.Y

11

22

1

N.Y

36

36

1

Fact:

April 15, 1469: Day of the Week

April 15, 1469 was the 105th day of the year 1469 in the Gregorian calendar. There were 260 days remaining until the end of the year. The day of the week was *Thursday*.

The day of the week for April 15, 1469 under the old Julian calendar was Saturday. Did you notice the difference with the Gregorian calendar?

2nd Oct, 1869

6088 days

1800 ⇒ 3

1868 ⇒ 4

+2

68 → 1

L.Y N.Y

17 51

34 51

6 + 2 = 8 → 1

J – 3 J – 2

F – 0 J – 3

M – 3 A – 3

A – 2 S – 2

M – 3+ O – 2

(A) Monday

(B) Saturday

(C) Wednesday

(D) Thursday

10th day ← 3rd Feb, 2025

MONDAY

2000 → 0

2024 → 2

24 → 2

L.Y N.Y

6 18

12 18

5 4 = 9

2 + 6 = 8

2024 → 2

J - 3

F - 3

6

Questionnaire:

2017 → 1

2018 → 1

2019 → 1

2020 → 2

2021 → 1

N.Y

2011 → 1

N.Y

2022

2012 → 2

2013 → 1

2014 → 1

2015 → 1

2016 → 2

[MCQ]

#Q. 2005 calendar is same as which of the given years?

A. 2009

B. 2016

C. 2011

D. 2015

2005 → 1

2011

2006 → 1

2007 → 1

2008 → 2

2009 → 1

2010 → 1

[MCQ]

#Q. When does 2016 calendar repeats?

A. 2027

B. 2040

C. ~~2021~~

D. 2044

2016 → 2

2017 → 1

2018 → 1

2019 → 1

2020 → 2

2016
+28
2044

$$\begin{array}{r} 2024 \\ +28 \\ \hline 2052 \end{array}$$

$$\begin{array}{r} 2004 \\ +28 \\ \hline 2032 \end{array}$$

$$\begin{array}{r} 2017 \\ +6 \\ \hline 2023 \end{array}$$

$$\begin{array}{r} 2012 \\ +28 \\ \hline 2040 \end{array}$$

L.Y. → 28 yrs

N.Y → 6 yrs

N.Y → 11 yrs

N.Y. → 11 yrs

L.Y. → 28 yrs

$$\begin{array}{r} 2005 \\ +6 \\ \hline 2011 \end{array}$$

2 mins Summary

Topic Calendars ~~& Clocks~~

① Date → Day?

② Calendar Repetition

3 → Conditional ?

THANK - YOU